

12.421/12.621 Lecture 9: Volume Scattering

Brent Minchew

Topics covered

- Fundamentals of volume scattering
- Heuristic scattering from vegetation
- Heuristic scattering from snow

1 Basics of volume scattering from heterogeneous media

- Distinction between surface and volume scattering
 - Surface scattering
 - * Reflectivity a function of
 1. dielectric constant of surface;
 2. tilting of surface relative to plane of incidence (i.e., long-wavelength ($\gg \lambda$) surface height variations).
 - * Directionality of scattering, only function of surface roughness.
 - Volume scattering
 - * Reflectivity function of
 1. dielectric discontinuities within the volume;
 2. density of dielectric heterogeneities.
 - * Directionality of scattering function of
 1. roughness of boundary surface;
 2. average dielectric constant of media;
 3. size of inclusions.
- How we think about and model volume scattering depends primarily on
 - size distribution of scatterers (relative to λ)
 - * large scatterers will be treated differently from small scatterers;
 - orientation of scatterers;
 - shape distribution of scatterers;
 - dielectric contrast between scatterers and background material.
- The key questions
 - What approximations are applicable? E.g.,
 - * large, randomly oriented cylinders (often good for vegetation);

- * small spherical scatterers (e.g., snow, atmosphere, etc.).
- Can we assume the volume layer acts as a single scatterer?
- Are losses dominated by absorption or scattering?
- Azimuthal symmetry?

2 Heuristic model for vegetation

2.1 Single-scattering model and scattering paths

- “Single-scattering model,” which means only one (or zero) bounces within the volume; can have up to 3 total bounces when counting surface scattering from the ground.
- Expected scattering types and paths (Figure 1).

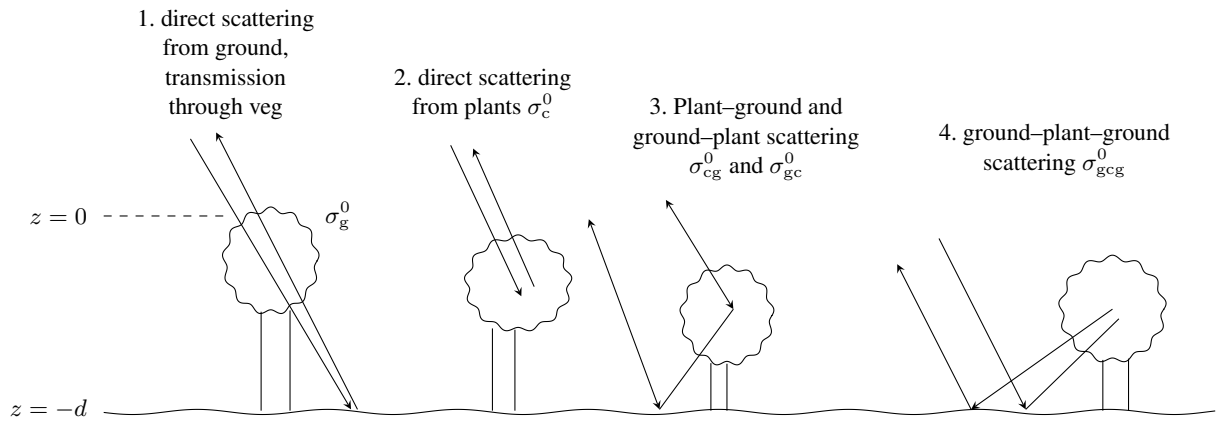


Figure 1: Single-scattering contributions to backscatter from a vegetation canopy of height d (cf. Ulaby and Long (2014), Fig. 11-3).

- Key point: top of vegetation ($z = 0$) is diffuse $\Rightarrow$ no scattering; important simplification that will not be valid for all volume scattering (e.g., snow is a volume scatterer with surface scattering component).

2.2 Extinction and transmissivity of the canopy

- To account for transmission through canopy, define p -polarized extinction coefficient (a.k.a. power attenuation factor)

$$\kappa_e^p = \kappa_a^p + \kappa_s^p \quad \rightarrow \quad \text{related to penetration depth} \rightarrow \delta_p = \frac{1}{\kappa_e}, \quad (1)$$

$\kappa_a^p \rightarrow$ absorption losses $\rightarrow \kappa_a^p = -2k_0 \text{Im}\{\sqrt{\epsilon}\}$ (cf. α , attenuation constant, from previous lectures),

$\kappa_s^p \rightarrow$ scattering losses.

In general, κ_e^p , κ_a^p , κ_s^p vary in space and time, particularly with z . Hereafter, assume κ_e^p , κ_a^p , κ_s^p are uniform.

- Now define transmissivity of canopy

$$\Upsilon_p = e^{-\kappa_e^p d / \cos \theta_i}. \quad (2)$$

2.3 Contribution 1: direct scattering from the ground

- Let us consider each scatterer defined in the previous cartoon (Figure 1).
- Recalling the previous lecture on surface scattering, we can expect a ground-scattered component in the absence of vegetation to be $\sigma_{s,pq}^0(\theta_i)$. With vegetation, we expect some attenuation due to absorption and scattering in the canopy

$$\Rightarrow \sigma_{g,pq}^0 = \Upsilon_p \Upsilon_q \sigma_{s,pq}^0, \quad p, q = h \text{ or } v. \quad (3)$$

2.4 Contribution 2: direct volume contribution

- There are many ways to model this component.
- By far the simplest is to treat as a “cloud”

$$\hat{\sigma}_{pq}^{\text{back}} = N_v \sigma_{pq}^{\text{back}}, \quad (4)$$

where

- $\hat{\sigma}_{pq}^{\text{back}} \equiv$ volume backscattering coefficient (m^{-1}),
- $\sigma_{pq}^{\text{back}} \equiv$ backscattering cross section of a single particle (m^2),
- $N_v \equiv$ number of “particles” per unit volume (m^{-3}).

- Visualize (Figure 2).

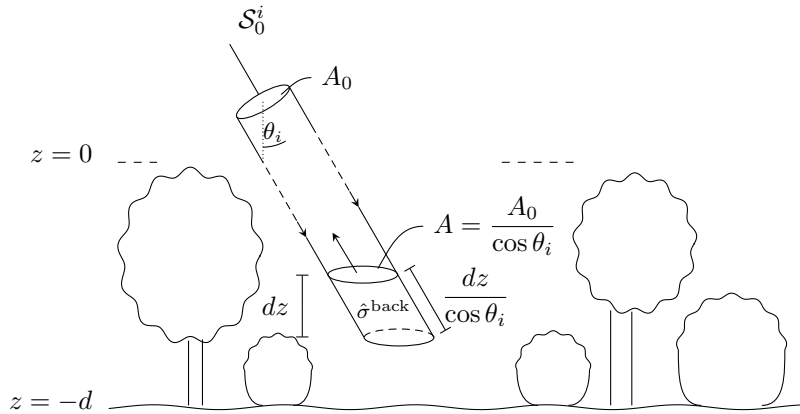


Figure 2: Direct backscatter by a differential volume of the canopy at depth z : a beam of cross section A_0 incident at θ_i illuminates the horizontal area $A = A_0 / \cos \theta_i$, and a layer of thickness dz is traversed along the slant length $dz / \cos \theta_i$, so the element has geometric volume $A_0 dz / \cos \theta_i = A dz$ (cf. Ulaby and Long (2014), Fig. 11-5).

- (q -polarized) power density (recall Poynting vector) across area A of a differential volume element within the canopy at depth $z < 0$

$$\mathcal{S}_q^i(z) = \mathcal{S}_0^i \cos \theta_i e^{\kappa_e^q z / \cos \theta_i} \left[\frac{\text{W}}{\text{m}^2} \right], \quad (5)$$

where $\mathcal{S}_0^i \equiv$ incident power density in free space.

- Backscattering cross section of differential volume

$$d\sigma_{pq}^{\text{back}} = \underbrace{\hat{\sigma}_{pq}^{\text{back}}}_{\text{backscattering cross section / unit volume}} \underbrace{\frac{A dz}{\cos \theta_i}}_{\text{volume of differential element}} [\text{m}^2]. \quad (6)$$

Here $A dz / \cos \theta_i$ is not the geometric volume of the element in Figure 2, which is $A dz = A_0 dz / \cos \theta_i$ (cross section A_0 times slant length $dz / \cos \theta_i$). The extra factor $1 / \cos \theta_i$ is compensated by the $\cos \theta_i$ in Eq. (5), which gives the power per unit horizontal area rather than the power density $\mathcal{S}_q^i / \cos \theta_i$ along the beam, so the product $\mathcal{S}_q^i d\sigma_{pq}^{\text{back}} = (\mathcal{S}_q^i / \cos \theta_i) \hat{\sigma}_{pq}^{\text{back}} A dz$ used below is correct and the results are unaffected.

- p -polarized power reradiated by differential volume, in the backscatter direction, after propagation from depth $z < 0$ to top of canopy ($z = 0$)

$$\begin{aligned} dP_p^{\text{rer}} &= \mathcal{S}_q^i(z) d\sigma_{pq}^{\text{back}} e^{\kappa_e^p z / \cos \theta_i} \\ &= \mathcal{S}_0^i A \hat{\sigma}_{pq}^{\text{back}} e^{(\kappa_e^p + \kappa_e^q)z / \cos \theta_i} dz. \end{aligned} \quad (7)$$

- Integrating dP_p^{rer} over the canopy depth

$$\begin{aligned} P_p^{\text{rer}} &= \mathcal{S}_0^i A \int_{-d}^0 \hat{\sigma}_{pq}^{\text{back}} e^{(\kappa_e^p + \kappa_e^q)z / \cos \theta_i} dz \\ &= \left\{ \frac{\mathcal{S}_0^i A \cos \theta_i \hat{\sigma}_{pq}^{\text{back}}}{\kappa_e^p + \kappa_e^q} \right\} \left[1 - e^{-(\kappa_e^p + \kappa_e^q)d / \cos \theta_i} \right] \\ &= \left\{ \frac{\mathcal{S}_0^i A \cos \theta_i \hat{\sigma}_{pq}^{\text{back}}}{\kappa_e^p + \kappa_e^q} \right\} [1 - \Upsilon_p \Upsilon_q]. \end{aligned} \quad (8)$$

- Defining the normalized radar cross section from the canopy as

$$\sigma_{c,pq}^0 = \frac{4\pi R^2}{A} \frac{\mathcal{S}_{r,p}}{\mathcal{S}_0^i}, \quad R \equiv \text{distance to antenna}, \quad (9)$$

where

$$\mathcal{S}_{r,p} = \frac{P_p^{\text{rer}}}{4\pi R^2}$$

is the power density at the receiving antenna, we have

$$\begin{aligned} \sigma_{c,pq}^0 &= \frac{\hat{\sigma}_{pq}^{\text{back}} \cos \theta_i}{\kappa_e^p + \kappa_e^q} (1 - \Upsilon_p \Upsilon_q) \\ &= \frac{\hat{\sigma}_{pq}^{\text{back}} \cos \theta_i}{\kappa_e^p + \kappa_e^q} \left(1 - e^{-(\kappa_e^p + \kappa_e^q)d \sec \theta_i} \right), \end{aligned} \quad (10)$$

where $\hat{\sigma}_{pq}^{\text{back}}$, κ_e^p and κ_e^q are all functions of the density and dielectric properties of the leaves and stems that make up the canopy.

2.5 Contribution 3: canopy-ground contributions

- Here we have reflection from ground then scattering from canopy and vice versa.
- Assume the ground is a quasi-specular reflector, meaning that we only consider reflection in the specular direction but account for diffuse scattering due to surface roughness in the reflectivity.
- Assume a very simple model

$$\Gamma_p^{\text{coh}} = \Gamma_p(\theta_i) e^{-4k_0^2 h^2 \cos^2 \theta_i}, \quad (11)$$

where

$\Gamma_p(\theta_i) \equiv |\rho_p|^2$, where $\rho_p \equiv$ Fresnel reflection coefficient,

$h \equiv$ RMS height of surface roughness.

- Sketch (Figure 3).

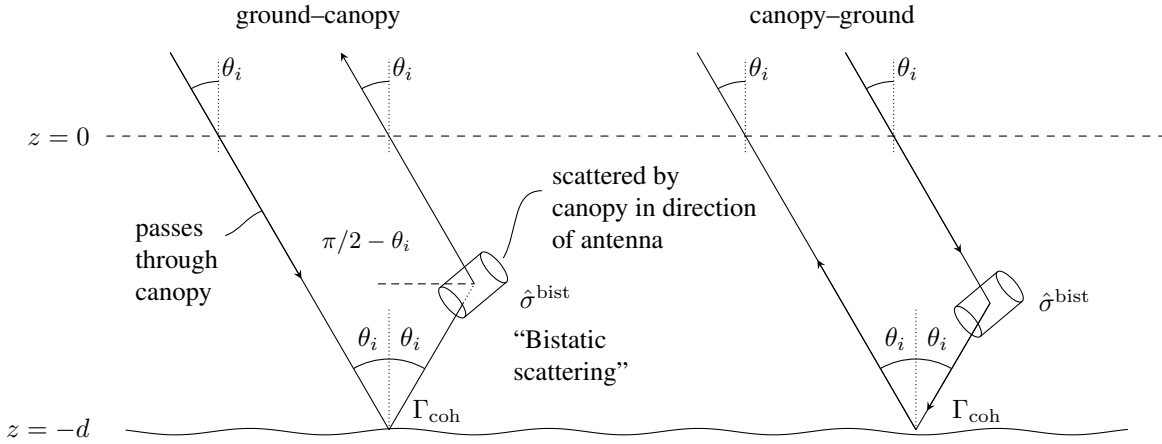


Figure 3: The two canopy-ground single-scattering contributions: ground-canopy (reflection by the ground, then bistatic scattering by the canopy toward the antenna) and canopy-ground (bistatic scattering by the canopy toward the ground, then reflection by the ground). The two follow the same path in reverse (cf. Ulaby and Long (2014), Fig. 11-6).

- Normalized radar cross sections

ground-canopy:

$$\sigma_{gc,pq}^0(\theta_i) = \hat{\sigma}_{pq}^{\text{bist}}(\pi/2 - \theta_i) \underbrace{d}_{\text{canopy thickness}} \Gamma_q^{\text{coh}} \Upsilon_q \Upsilon_p, \quad (12)$$

canopy-ground:

$$\sigma_{cg,pq}^0(\theta_i) = \hat{\sigma}_{pq}^{\text{bist}}(\pi/2 - \theta_i) d \Gamma_p^{\text{coh}} \Upsilon_p \Upsilon_q. \quad (13)$$

- The total canopy-ground contribution

$$\begin{aligned} \sigma_{cgt,pq}^0 &= \sigma_{gc,pq}^0 + \sigma_{cg,pq}^0 \\ &= \hat{\sigma}_{pq}^{\text{bist}} d \left(\Gamma_p^{\text{coh}} + \Gamma_q^{\text{coh}} \right) \Upsilon_p \Upsilon_q. \end{aligned} \quad (14)$$

- In the case of co-polarized scattering ($p = q$), we have

$$\sigma_{\text{cgt},pp}^0 = 2\hat{\sigma}_{pp}^{\text{bist}} d \Gamma_p^{\text{coh}} \Upsilon_p^2. \quad (15)$$

- However, this (above) is not quite correct because we added the powers (for simplicity) rather than taking the power of the total $\mathbf{E}$ field. In other words, we undertook incoherent addition of fields. Since ground–canopy and canopy–ground follow the same path in reverse, we should expect equivalent phase $\Rightarrow$ $\mathbf{E}$ fields add coherently. Can correct by simply multiplying by factor of two

$$\Rightarrow \quad \boxed{\sigma_{\text{cgt},pp}^0 = 4\hat{\sigma}_{pp}^{\text{bist}} d \Gamma_p^{\text{coh}} \Upsilon_p^2 \quad \text{for } p = h \text{ or } v} \quad (16)$$

2.6 Contribution 4: ground–canopy–ground contribution

- This is very simple: 2-way transmission, 2 bounces off the ground, and backscatter from the canopy $\sigma_{c,pq}^0(\theta_i)$

$$\begin{aligned} \Rightarrow \quad \sigma_{\text{gcg},pq}^0 &= \Upsilon_p \Upsilon_q \Gamma_p^{\text{coh}} \Gamma_q^{\text{coh}} \sigma_{c,pq}^0 \\ &= \frac{\hat{\sigma}_{pq}^{\text{back}} \cos \theta_i}{\kappa_e^p + \kappa_e^q} \Gamma_p^{\text{coh}} \Gamma_q^{\text{coh}} (\Upsilon_p \Upsilon_q - \Upsilon_p^2 \Upsilon_q^2). \end{aligned} \quad (17)$$

2.7 Putting it all together

- ★ Putting it all together, the total normalized radar cross section is

$$\begin{aligned} \sigma_{pq}^0 &= \sigma_{\text{g},pq}^0 + \sigma_{\text{c},pq}^0 + \sigma_{\text{cgt},pq}^0 + \sigma_{\text{gcg},pq}^0 \\ &= \Upsilon_p \Upsilon_q \sigma_{\text{s},pq}^0(\theta_i) + \frac{\hat{\sigma}_{pq}^{\text{back}} \cos \theta_i}{\kappa_e^p + \kappa_e^q} (1 - \Upsilon_p \Upsilon_q) \left(1 + \Gamma_p^{\text{coh}} \Gamma_q^{\text{coh}} \Upsilon_p \Upsilon_q\right) \\ &\quad + c_{\text{add}} \hat{\sigma}_{pq}^{\text{bist}} d \left(\Gamma_p^{\text{coh}} + \Gamma_q^{\text{coh}}\right) \Upsilon_p \Upsilon_q, \end{aligned} \quad (18)$$

where

$$c_{\text{add}} = \begin{cases} 1 & p \neq q \quad (\text{cross-pol, incoherent addition}), \\ 2 & p = q \quad (\text{co-pol, coherent addition}). \end{cases}$$

In general, the terms $\sigma_{\text{s},pq}^0$, $\hat{\sigma}_{pq}^{\text{back}}$, $\hat{\sigma}_{pq}^{\text{bist}}$, $\kappa_e^p (\Rightarrow \Upsilon_p)$, $\kappa_e^q (\Rightarrow \Upsilon_q)$, Γ_p^{coh} , Γ_q^{coh} , and d are not known a priori.

2.8 Simplifications

- Canopy of negligible thickness:

$$\lim_{d \rightarrow 0} \sigma_{pq}^0 = \sigma_{\text{s},pq}^0 \quad (\text{pure surface scattering from the ground}). \quad (19)$$

- Very thick and/or dense canopy:

$$\Upsilon_p, \Upsilon_q \rightarrow 0 \quad \Rightarrow \quad \sigma_{pq}^0 \rightarrow \frac{\hat{\sigma}_{pq}^{\text{back}} \cos \theta_i}{\kappa_e^p + \kappa_e^q}. \quad (20)$$

- Isotropic scatterers:

$$\hat{\sigma}^{\text{back}} = \hat{\sigma}^{\text{bist}} = \kappa_s = \kappa_e + 2k_0 \text{Im}\{\sqrt{\varepsilon}\} \quad (\text{note } \kappa_e = \kappa_s + \kappa_a, \kappa_a = -2k_0 \text{Im}\{\sqrt{\varepsilon}\}) \quad (21)$$

and $\Upsilon_p = \Upsilon_q$

$$\Rightarrow \sigma_{pp}^0 = \Upsilon^2 \sigma_{s,pp}^0(\theta_i) + \frac{\kappa_s}{\kappa_e} \frac{\cos \theta_i}{2} (1 - \Upsilon^2) \left(1 + (\Gamma^{\text{coh}})^2 \Upsilon^2\right) + 4\kappa_s d \Gamma^{\text{coh}} \Upsilon^2, \quad (22)$$

where $\Upsilon \equiv \Upsilon_p$ and $\Gamma^{\text{coh}} \equiv \Gamma_p^{\text{coh}}$.

- Rayleigh scatterers (spheres with radius $\ll \lambda$):

$$\hat{\sigma}^{\text{back}} = \hat{\sigma}^{\text{bist}} = \frac{3}{2} \kappa_s \quad (\text{bistatic value: } hh; \text{ see below for } vv) \quad (23)$$

$$\Rightarrow \sigma_{pp}^0(\theta_i) = \Upsilon^2 \sigma_{s,pp}^0 + \frac{3}{4} \frac{\kappa_s}{\kappa_e} \cos \theta_i (1 - \Upsilon^2) \left(1 + (\Gamma^{\text{coh}})^2 \Upsilon^2\right) + 3c_{\text{add}} \kappa_s d \Gamma^{\text{coh}} \Upsilon^2. \quad (24)$$

(For vv , the dipole scattering pattern reduces the bistatic value in the specular geometry of Figure 3, where the scattering angle is $2\theta_i$, to $\hat{\sigma}^{\text{bist}} = \frac{3}{2} \kappa_s \cos^2 2\theta_i$, so the last term of Eq. (24) becomes $3c_{\text{add}} \kappa_s d \Gamma^{\text{coh}} \cos^2(2\theta_i) \Upsilon^2$; the backscatter value $\frac{3}{2} \kappa_s$ holds for both polarizations.)

- General considerations for models of $\hat{\sigma}^{\text{back}}$ and $\hat{\sigma}^{\text{bist}}$:
 - density of scatterers;
 - size of scatterers relative to EM wavelength;
 - if size $\gtrsim \lambda$, shape and orientation of scatterers.

3 Heuristic model of scattering from a snow layer of depth d

- The volume scattering will be identical to the vegetation model, but now we must account for reflection and transmission at the air–snow interface.
- Straightforward extension of previous model

$$\sigma_{pq}^0(\theta_i) = \mathbb{T}_p \mathbb{T}_q \left(\sigma_{pq}^0|_{\text{veg}} \right) + \sigma_{\text{as},pq}^0, \quad (25)$$

where

$\mathbb{T}_p$ and $\mathbb{T}_q$ are transmissivity of air–snow interface (at θ_i),

$\sigma_{pq}^0|_{\text{veg}}$ is the model derived previously for vegetation scattering (Eq. (18), evaluated at the angle of refraction θ_t into the snow, with $\sigma_{s,pq}^0$ and Γ_p^{coh} those of the snow–ground interface),

$\sigma_{\text{as},pq}^0$ is surface scattering from air–snow interface.

4 Summary

- Volume scattering involves scattering from dielectric discontinuities within a volume.
- Reflectivity of a volume is function of
 - dielectric discontinuities within the volume;

- density of dielectric heterogeneities.
- Directionality of scattering
 - roughness of boundary;
 - average dielectric constant of media;
 - size of inclusions.
- Modeling of volume scattering depends on
 - size distribution of scatterers (relative to λ);
 - orientation of scatterers;
 - shape of scatterers;
 - dielectric contrast.
- Multiple scattering contributions from volume. E.g., vegetation
 - direct ground with 2-way transmission through volume;
 - direct canopy contribution;
 - ground–canopy and canopy–ground;
 - ground–canopy–ground.
- For volumes with nondiffuse surface, must consider transmission through and scattering from interface.

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Geometry and general</i>			
λ	EM wavelength	m	–
k_0	free-space wavenumber, $2\pi/\lambda_0$	rad/m	–
λ_0	free-space wavelength	m	–
θ_i	angle of incidence	rad	–
θ_t	angle of refraction (into the snow)	rad	–
z	vertical coordinate ($z = 0$ top of the canopy, $z = -d$ ground)	m	–
d	canopy (or snow-layer) thickness	m	–
p, q	receive and transmit polarizations	–	h or v
R	distance to the antenna	m	–
A_0	cross-sectional area of the incident beam	m ²	–
A	horizontal area illuminated at depth z	m ²	$A_0 / \cos \theta_i$
<i>Extinction and transmission</i>			
κ_e^p	extinction coefficient (p polarization)	Np/m	$\kappa_a^p + \kappa_s^p$
κ_a^p	absorption coefficient	Np/m	$-2k_0 \text{Im}\{\sqrt{\varepsilon}\}$
κ_s^p	scattering coefficient	Np/m	–
ε	(effective) dielectric constant of the canopy medium	dimensionless	–
α	attenuation constant (Lecture 2)	Np/m	$\kappa_a = 2\alpha$
δ_p	penetration depth	m	$1/\kappa_e$
Υ_p	one-way transmissivity of the canopy	dimensionless	$e^{-\kappa_e^p d / \cos \theta_i}$
$\mathbb{T}_p$	transmissivity of the air–snow interface	dimensionless	0–1
<i>Scattering</i>			
σ_{pq}^0	total normalized radar cross section	dimensionless	–
$\sigma_{s,pq}^0$	backscattering coefficient of the bare ground surface	dimensionless	–
$\sigma_{g,pq}^0, \sigma_{c,pq}^0$	direct ground and direct canopy contributions	dimensionless	–
$\sigma_{gc,pq}^0, \sigma_{cg,pq}^0$	ground–canopy and canopy–ground contributions	dimensionless	–
$\sigma_{cgt,pq}^0$	total canopy–ground contribution	dimensionless	$\sigma_{gc,pq}^0 + \sigma_{cg,pq}^0$
$\sigma_{gcg,pq}^0$	ground–canopy–ground contribution	dimensionless	–
$\sigma_{as,pq}^0$	backscattering coefficient of the air–snow interface	dimensionless	–
$\hat{\sigma}_{pq}^{\text{back}}$	volume backscattering coefficient	m ^{–1}	$N_v \sigma_{pq}^{\text{back}}$
$\hat{\sigma}_{pq}^{\text{bist}}$	volume bistatic scattering coefficient	m ^{–1}	–
$\sigma_{pq}^{\text{back}}$	backscattering cross section of a single particle	m ²	–
N_v	number of particles per unit volume	m ^{–3}	–
$d\sigma_{pq}^{\text{back}}$	backscattering cross section of a differential volume	m ²	–

Symbol	Description	Units	Value
c_{add}	coherent-addition factor	dimensionless	1 ($p \neq q$), 2 ($p = q$)
<i>Power</i>			
S_0^i	incident power density in free space	W/m ²	–
$S_q^i(z)$	q -polarized power density across A at depth z	W/m ²	–
$S_{r,p}$	p -polarized power density at the receiving antenna	W/m ²	$P_p^{\text{rer}} / (4\pi R^2)$
P_p^{rer}	p -polarized power reradiated by the canopy	W	–
<i>Ground reflection</i>			
Γ_p	Fresnel reflectivity of the ground, $ \rho_p ^2$	dimensionless	0–1
Γ_p^{coh}	coherent reflectivity of the rough ground	dimensionless	$\Gamma_p e^{-4k_0^2 h^2 \cos^2 \theta_i}$
$\Upsilon, \Gamma^{\text{coh}}$	shorthand for $\Upsilon_p, \Gamma_p^{\text{coh}}$ in co-polarized forms (Eqs. 22–24)	dimensionless	–
ρ_p	Fresnel reflection coefficient	dimensionless	–
h	RMS height of the ground roughness	m	–